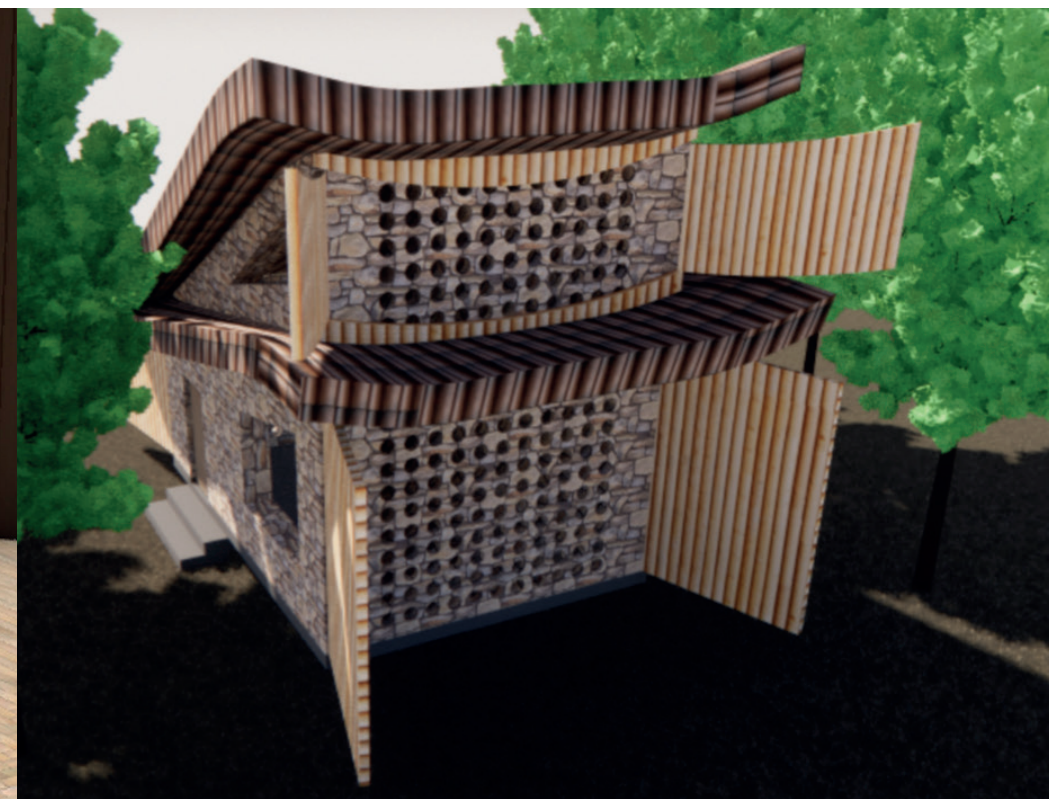


PORTFOLIO

ARCHINEERING 1



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TECHNICAL UNIVERSITY DELFT
NOVEMBER 2021

Porfolio minor Archineering 1

Analasis:

Konstantin Melnikow's house and studio

House of Constantijn Huygens

Villa Vals

Canvas house

Charles Moore's house

Huis hardglas

Alderton house

Projects:

Tiny house

CNC chair

Climate adaptive tiny house

November 2021

Elisa van Klink

TU Delft

Bachelor Bouwkunde

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Analysis – Konstatin Melnikow's house

4

1. Konstatin Melnikow



2. Birdseye view



3. Inside view

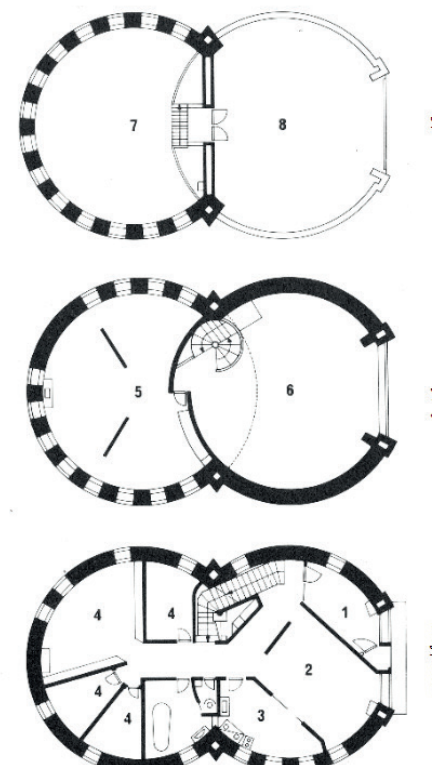
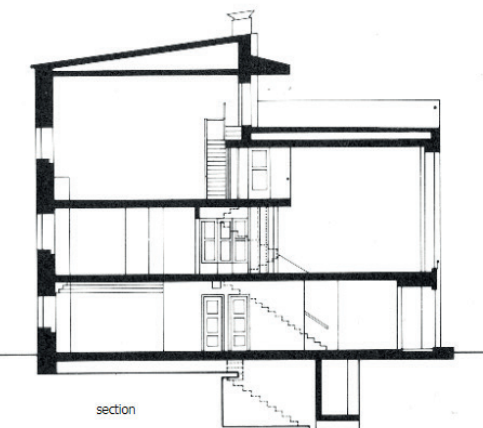


Introduction

Melnikov's house is a residential building located on Krivoarbatsky Lane in the Arbat district Moscow Russia. The house, which was finished in 1929, is a good representation of the Russian Avant Garde that took place in the 1920's. The design was and still is a very unconventional as in that era uniformity was the preferable outlook a building had to give.

The main concept of the building are the two interlocking cylinders that form the buildings volume. The reason the house had been given the shape was because Melnikov thought this would save material. One other iconic feature of the house are the hexagonal windows in one of the cylinders illuminating the inside from all around. The shapes were a direct result of the construction behind it because of the honeycomb lattice work made from local bricks and finished with white plaster. This structure uses a minimal amount of material while still having a strong foundation for the building.

4. Plans, sections and construction



5. Front door

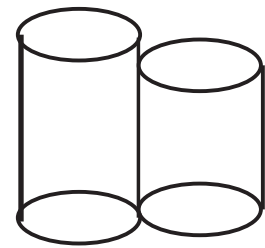


- 1: Main Moscow architects: Melnikov, Shchusev, Schechtel, Iofan. [2021]. Idea Guide. Geraadpleegd op 4 november 2021, van https://ideaguide.ru/en/moscow/moscow_architects
- 2 and 4: Gegeven tekeningen
- 3 and 5: Emden, C. [2015, 20 augustus]. Konstantin Stepanovich Melnikov, Cemal Emden · Melnikov House. Divisare. Geraadpleegd op 4 november 2021, van <https://divisare.com/projects/296176-konstantin-stepanovich-melnikov-cemal-emden-melnikov-house>
- Tekst: Winstanley, T. [2021, 20 augustus]. AD Classics: Melnikov House / Konstantin Melnikov. ArchDaily. Geraadpleegd op 5 november 2021, van <https://www.archdaily.com/151567/ad-classics-melnikov-house-konstantin-melnikov>

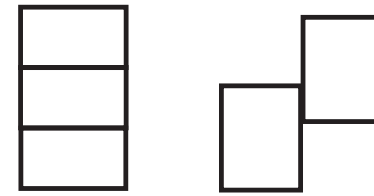
Analysis – Konstantin Melnikow's house

5

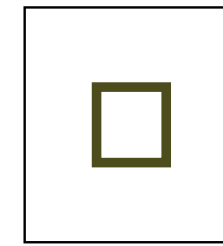
Form & space



General shape

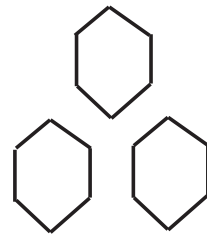


Stacked and leveled spaces

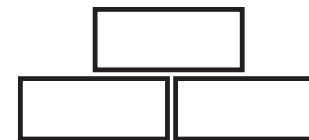


Situated in the middle of a plot

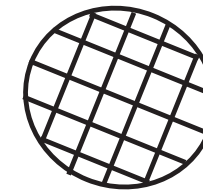
Materialisation
& construction



Honeycomb structure



Bricks (hidden)

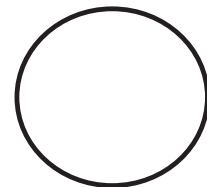


'Waffle' floors

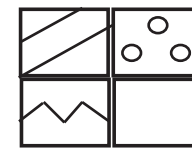


Smooth white outlook (plaster) --> hidden construction

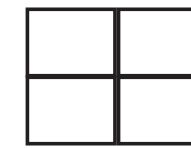
Functionality



Circular floorplan

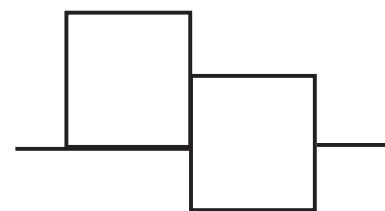


Distinct functions

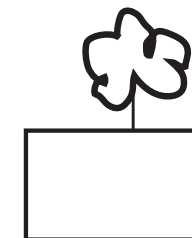


Closed floorplan

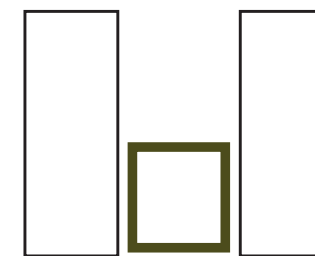
Urban context



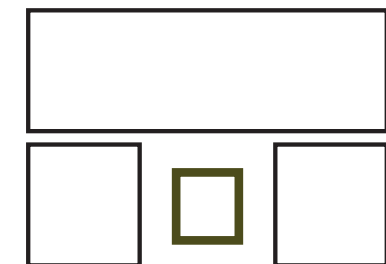
One volume lower than the other



Terrace on top of building



Hidden between higher buildings

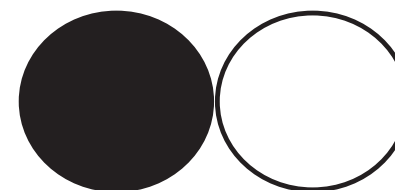


locked in

Social context



Residential building for one family



Private/ open area

Analysis – House of Constantijn Huygens

6

1. Constantijn Huygens



2. birdseye view



3. Entrance



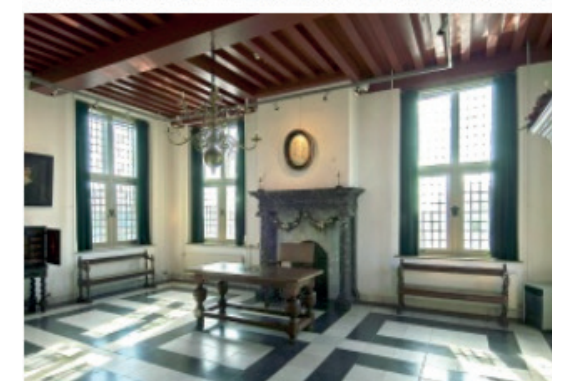
Introduction

The Huygens's residence, which is currently used as a museum, was designed by Christiaan Huygens himself in Voorburg the Netherlands 1642. The concept of the building was based on the concepts of the Architect Marcus and his drawings of the Vitruvian man. Based on the idea of the human shape and the harmony of the Vitruvian man, Huygens created a house with set dimensions and looking identical on each side but for the entrance. The house itself representing the head, and the garden the body. Another main feature is the placement of the building, it being partly under water seeming to float above it.

4. Plans, sections and construction



5. Inside view



1: Wikipedia contributors. (2021, 31 juli). Constantijn Huygens. Wikipedia. Geraadpleegd op 4 november 2021, van https://en.wikipedia.org/wiki/Constantijn_Huygens

2: CDA. [z.d.]. Tuin van Huygens' Hofwijck moet behouden blijven! CDA - Zuid-Holland. Geraadpleegd op 4 november 2021, van <https://www.cda.nl/zuid-holland/actueel/nieuws/tuin-van-huygens-hofwijck-moet-behouden-blijven>

3: Huygens Hofwijck. (2015). Buitenplaats I hofwijck.nl. Geraadpleegd op 4 november 2021, van <https://www.hofwijck.nl/nl/buitenplaats>

4 and 5: Gegeven tekeningen

Analysis – House of Constantijn Huygens

7

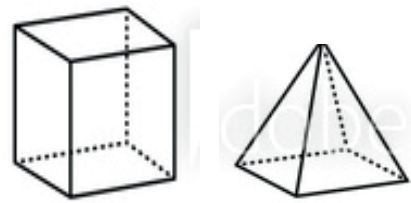
Form & space

Materialisation
& construction

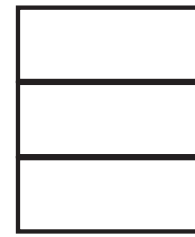
Functionality

Urban context

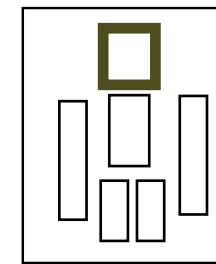
Social context



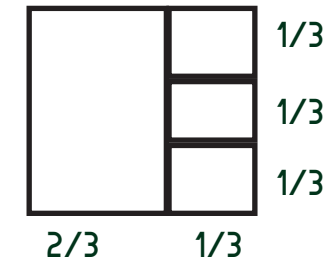
General shapes



Stacked spaces



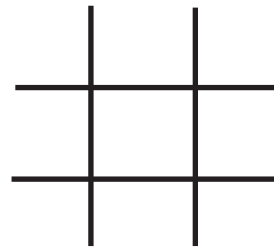
Situated at the top
of the plot
(the head)



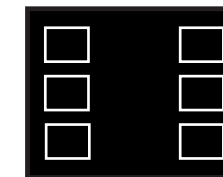
Distinct proportions



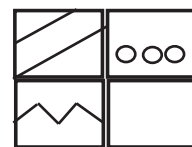
Bricks



Linear grid divided in
threes



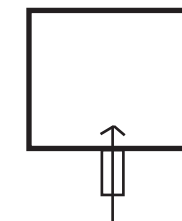
Closed facade



Distinct functions



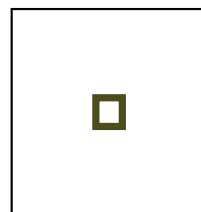
Square floorplan



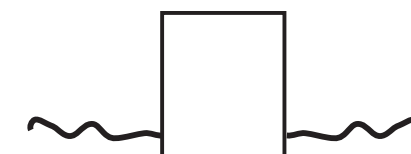
One entrance



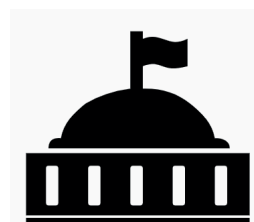
Closed floorplan



Small house on a
large plot



Partly under the
water



Monument and museum



Used to be vacation home

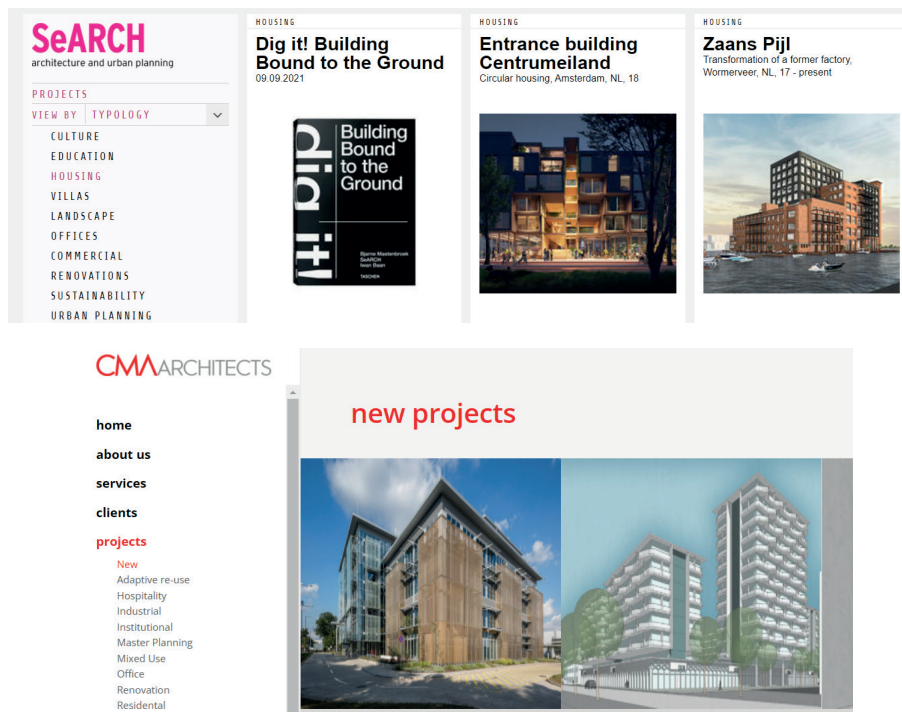


Closed off from
outside people

Analysis – Villa Vals

8

1. CMA and SeARCH



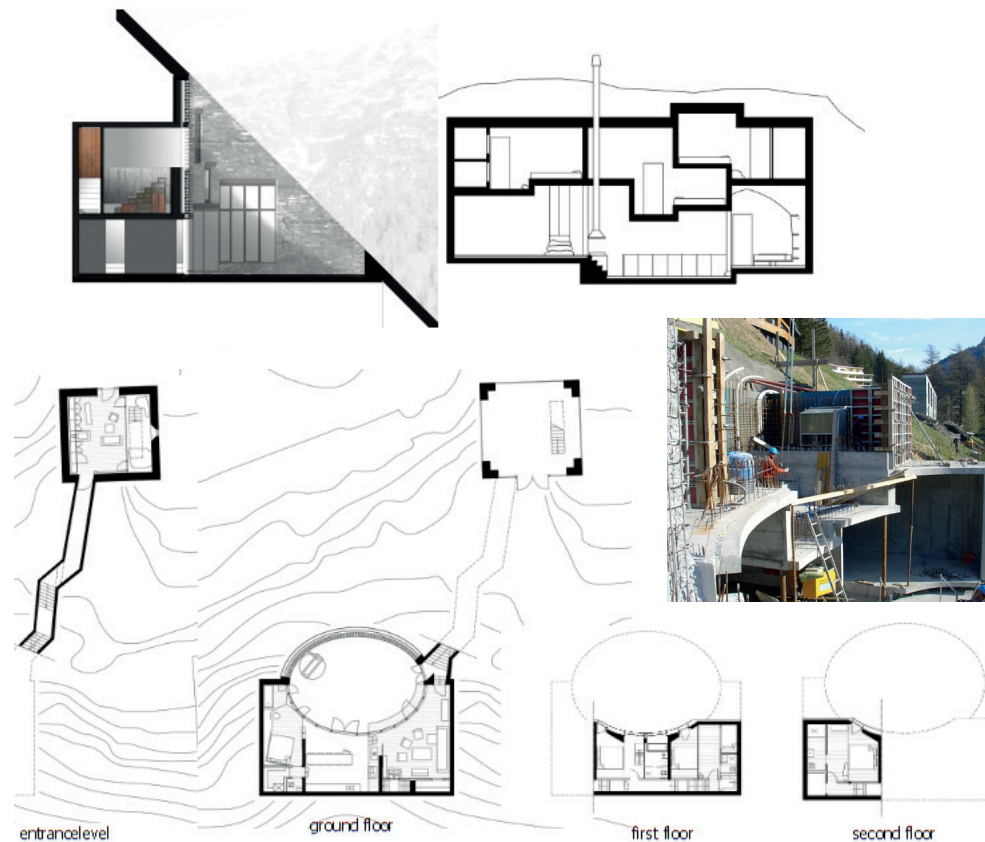
Introduction

Villa Vals is situated in Vals Switzerland and was built in 2009 by two architect firms CMA and SeARCH. Their main concept was to completely integrate the villa into the landscape to avoid disturb the unspoiled nature views from the thermal baths of Vals. The only giveaway is the rounded terrace with a slanted barrier around it that had been created to be able to go inside and enjoy the outside view. The entrance however is further down in an old shed (Graubünder shed). Once you enter the house every room is under the ground and yet lit by daylight as the façade mostly consists of windows catching every ray of sun.

2. Entrance to terras and house



3. Plans, sections and construction



4. Inside view



- 1: CMA. (2017). CMA architects. Geraadpleegd op 4 november 2021, van <http://cma.hu/cma/projects/new/> and SeARCH. (z.d.). SeARCH. Geraadpleegd op 4 november 2021, van <https://www.search.nl/>
- 2: Villa Vals. (2021). Villa Vals. Geraadpleegd op 4 november 2021, van <https://www.villavals.ch/> and gegeven tekeningen
- 3 en 4: Gegeven tekeningen
- Tekst: Saieh, N. (2021, 7 oktober). Villa Vals / SeARCH + CMA. ArchDaily. Geraadpleegd op 5 november 2021, van <https://www.archdaily.com/43187/villa-vals-se-arch-cma>

Analysis – Villa Vals

9

Form & space

Materialisation
& construction

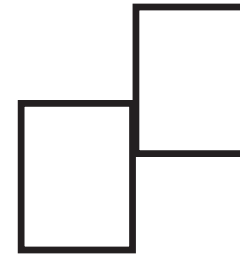
Functionality

Urban context

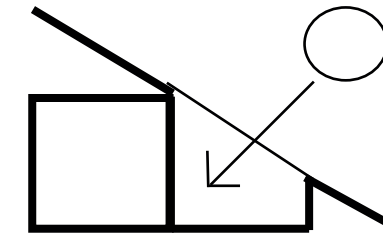
Social context



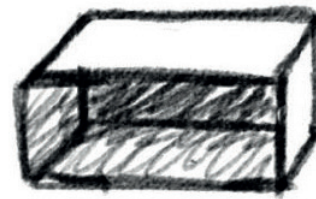
General shape



Different leveled spaces



Slanted walls on terrace to match slope and get enough sun



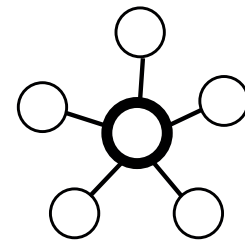
Open rectangle



Open and closed facade



Local/ rural materials in harmony with the environment



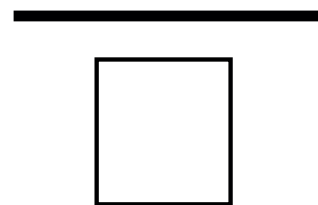
One central space with entrances to every room



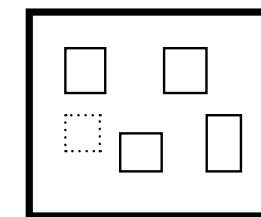
Distinct functions



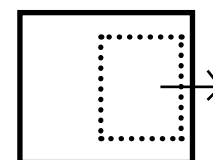
Closed floorplan



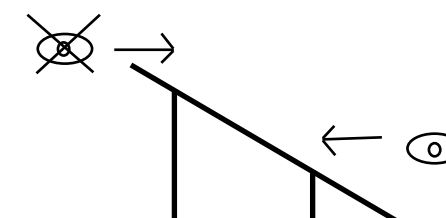
Hidden under ground



Hidden from the town



Open towards the outside and yet enclosed by walls



Only visible from a certain angle

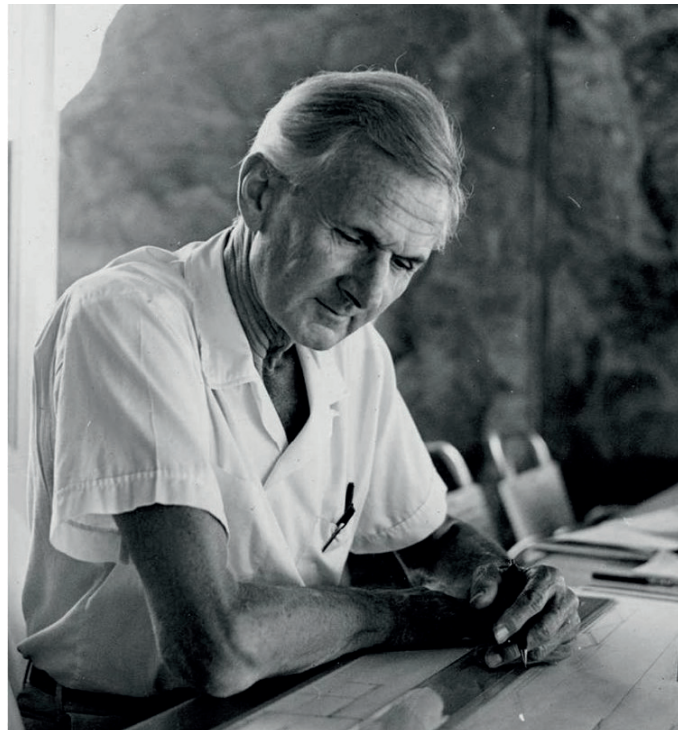


Residential building for one family

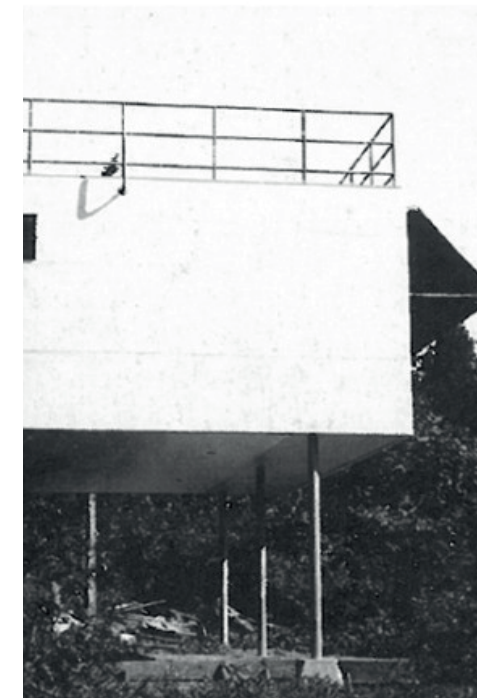
Analysis – Canvas house

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1. Albert Frey



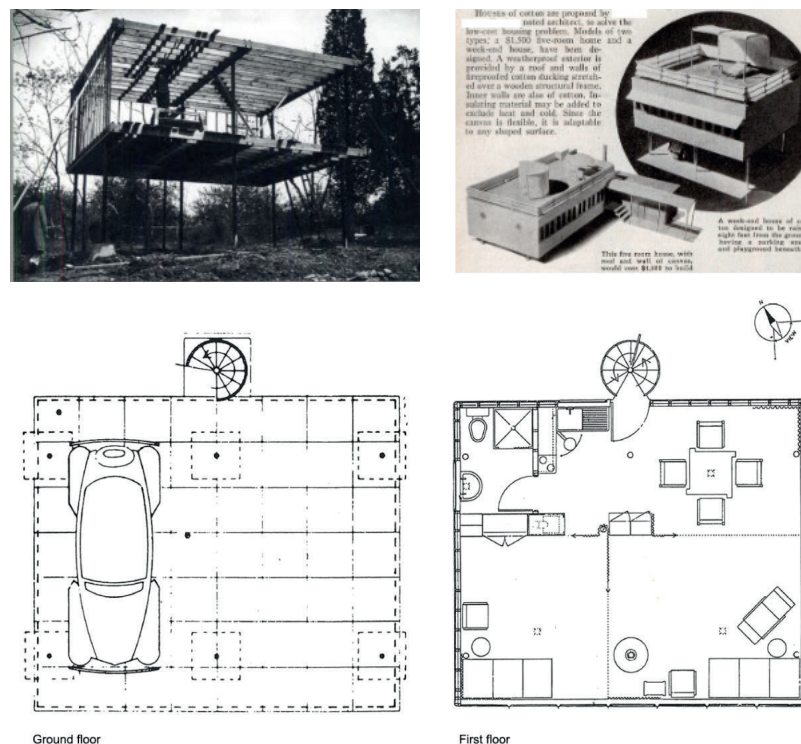
2. Outside view



Introduction

The canvas house, designed by Albert Frey in 1933 in Long Island New York, has its name from the somewhat odd material choice of the outer façade. The façade is namely made from canvas. Marine canvas stretched over a redwood frame to be exact. The cotton used in the exterior also help insulate the building and was very easy to apply since it was very easy to adapt to any surface. The house served as a weekend house and had a very open and flexible interior, raised high above the ground so people were able to stall their car under the building.

3. Plans, sections and construction



4. Inside view

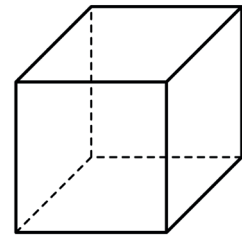


- 1: Henderson, R. [2021, 5 oktober]. Albert Frey: The Visionary Palm Springs Architect. Home. Geraadpleegd op 4 november 2021, van <https://www.atomic-ranch.com/architecture-design/albert-frey-visionary-palm-springs-architect/>
- 2: Elarafrizenwalden. [z.d.]. -. Tumblr. Geraadpleegd op 4 november 2021, van <https://elarafrizenwalden.tumblr.com/post/113008154770/kocher-canvas-weekend-house-demolished/embed> en gegeven tekeningen
- 3 en 4: Gegeven tekeningen
- Tekst: Gegeven artikel

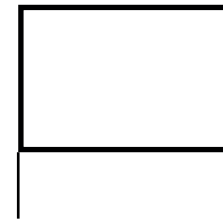
Analysis – Canvas house

11

Form & space

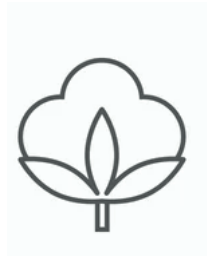


General shape

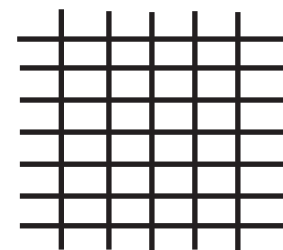


Lifted from the ground

Materialisation
& construction



Cotton

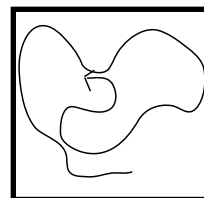


Wooden grid

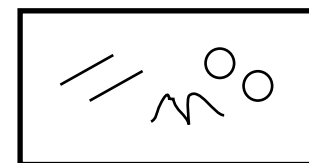


Open en gesloten facade

Fuctionality



Open floorplan



Every fuction on one floor,
no distinct seperation

Urban context



Lifted off the ground

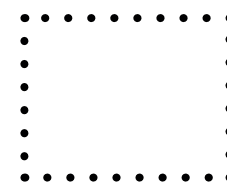


Green and open area

Social context



Weekend home



Open to the outside

Analysis – Charles Moore's house

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1. Charles Moore



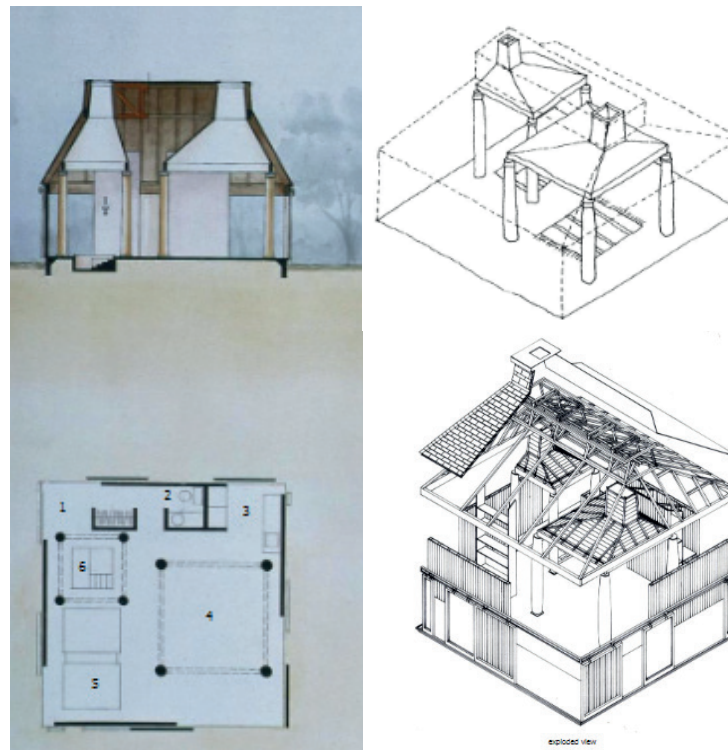
2. Outside view



Introduction

Charles Moore's house, or also known as the Orinda house, is designed by Charles himself in 1962 in San Francisco. With this project he wanted to experiment to recreate different tools of architectural history which would create an unconventional array of spaces. Making the house look more like archetypes such as a primitive hut. Another striking feature of the house are the barn like doors on the sides of the building that can be slid open to get more light in and open the space towards the outside forestry area. The roof of the building is held up by numerous techniques one of which are the eight columns who carry two smaller roof like structures underneath the main roof that brings everything together. There are also very little walls and the space is as much open on the inside as it is to the outside.

3. Plans, sections and construction



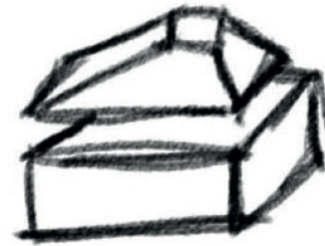
4. Inside view



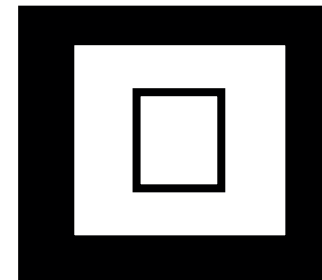
Analysis – Charles Moore's house

13

Form & space

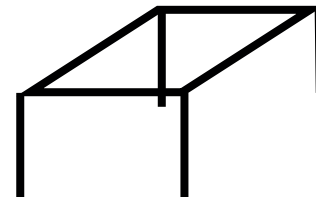


General shape



In an open space in the forest

Materialisation & construction



'Dak op poten'

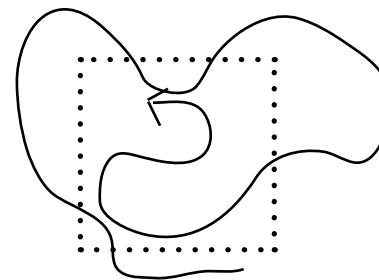


Open and closed facade
Can totally open/ fully close

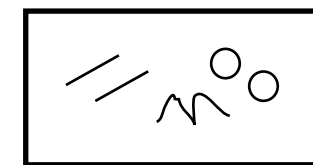


Use of local wood

Functionality



Open floorplan
even to the outside



Every function in one space,
no distinct separation

Urban context

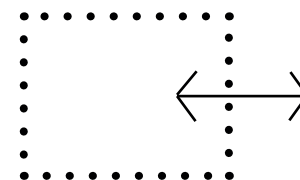


Slightly lifted off the ground

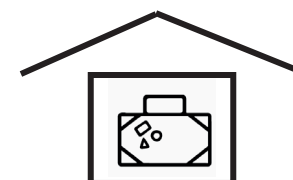


Green and foresty area

Social context



Inside is outside

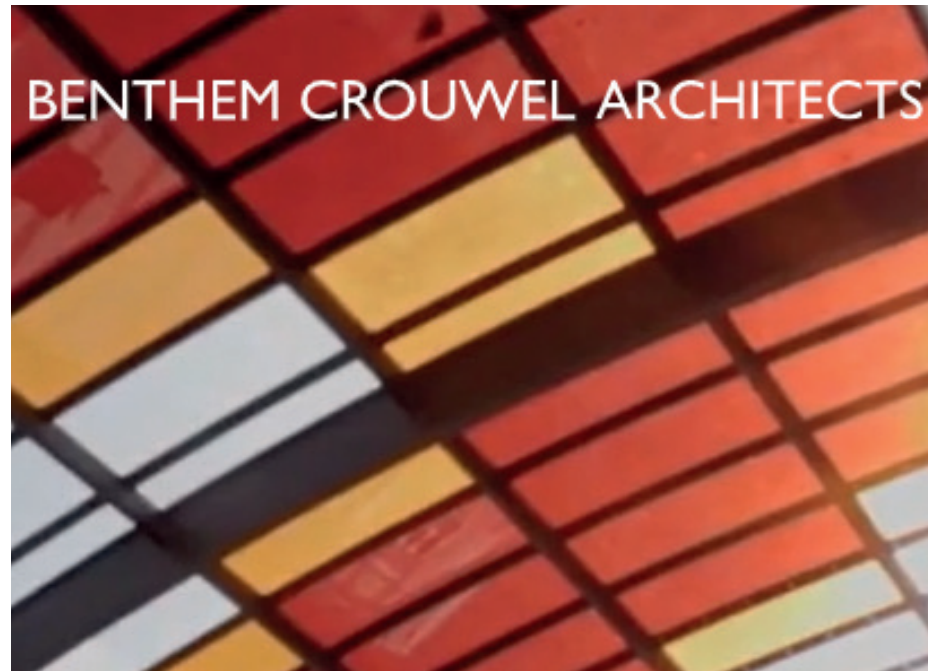


Vacation home

Analysis – House 'The Fantasy'

14

1. Benthem & Crouwel architects



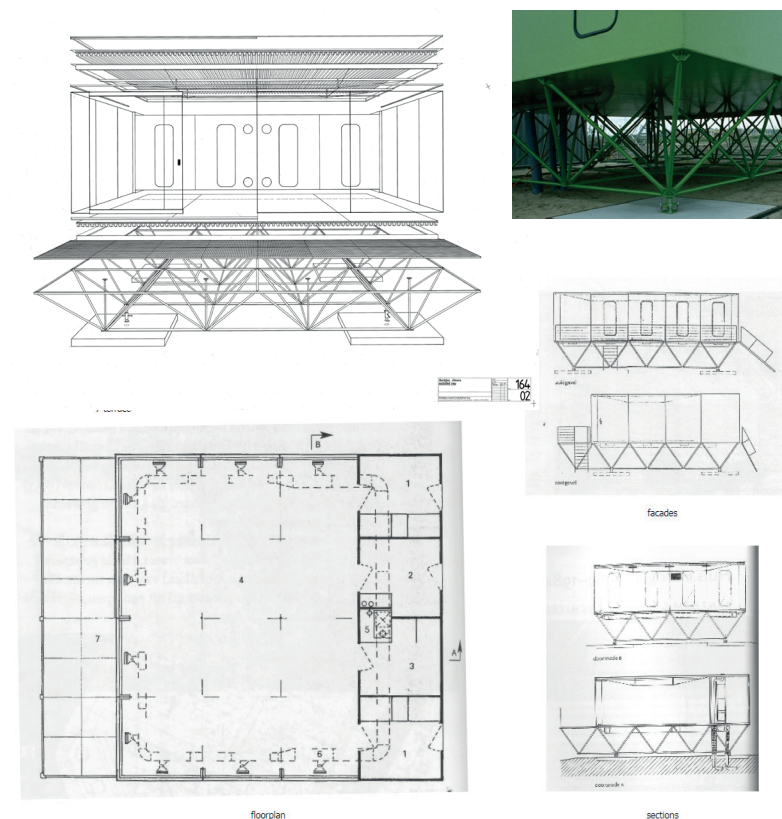
2. Outside view



Introduction

The Fantasy, designed by the Benthem Crouwel architects was created for a competition 'the unusual homes' in Almere. The architects were allowed to build/ design a house that didn't had to follow current building regulations. The building themselves also had to be dismantlable as they were only allowed to loan the property, they would be created on for five years. The most notable features of the house are of course the green steel foundation under the house and the huge glass façade. This glass façade can even be opened to gain even more floor space towards the outside, while in the green steel box you can also have some more privacy.

3. Plans, sections and construction



4. Inside view

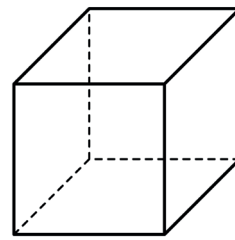


1, 2 en 4: Benthem Crouwel Architects. [z.d.]. Benthem Crouwel – Beyondarchitecture. Benthem Crouwel. Geraadpleegd op 5 november 2021, van <https://www.benthemcrouwel.com/>
2 en 3: Gegeven tekeningen
3: Octatube. (2021, 16 juli). Douane-emplacementen. Geraadpleegd op 5 november 2021, van https://www.octatube.nl/nl_NL/project-item.html/projectitem/146-hard-glas
Tekst: Benthem Crouwel. (2020). Benthem Crouwel – House De Fantasie. Geraadpleegd op 5 november 2021, van <https://www.benthemcrouwel.com/projects/house-de-fantasie>

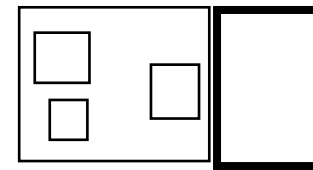
Analysis – House 'The Fantasy'

15

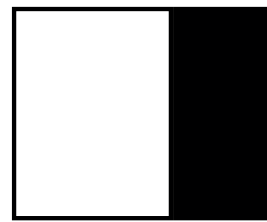
Form & space



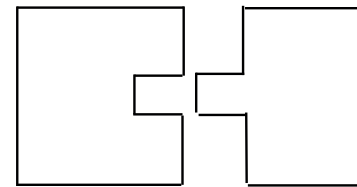
General shape



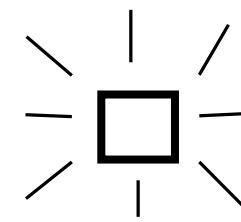
Modular plan and fixed plan



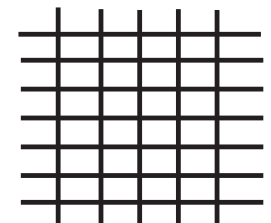
Open glass box and the closed metal box



Dismantable

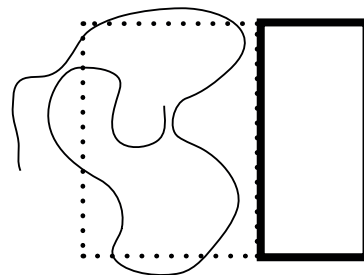


Visible construction

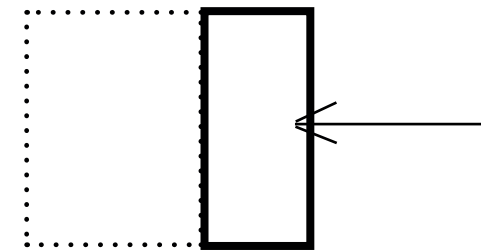


Steel grid

Materialisation
& construction

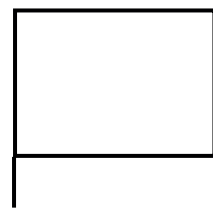


Open floorplan and closed floorplan

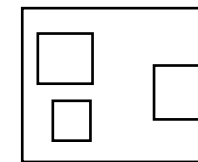


One entrance to get to the house at the closed area

Fuctionality

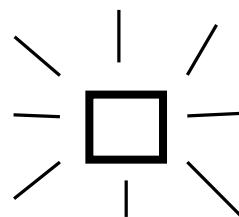


House is elevated from the ground

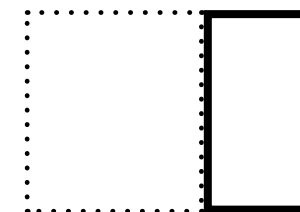


modular and can be placed everywhere

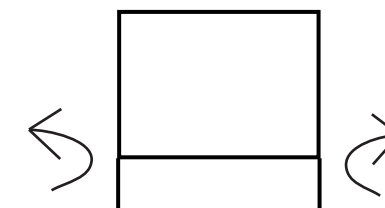
Urban context



Stands out from it's enviroment



Private and not private area



Outside people can't easily get to the house

Social context

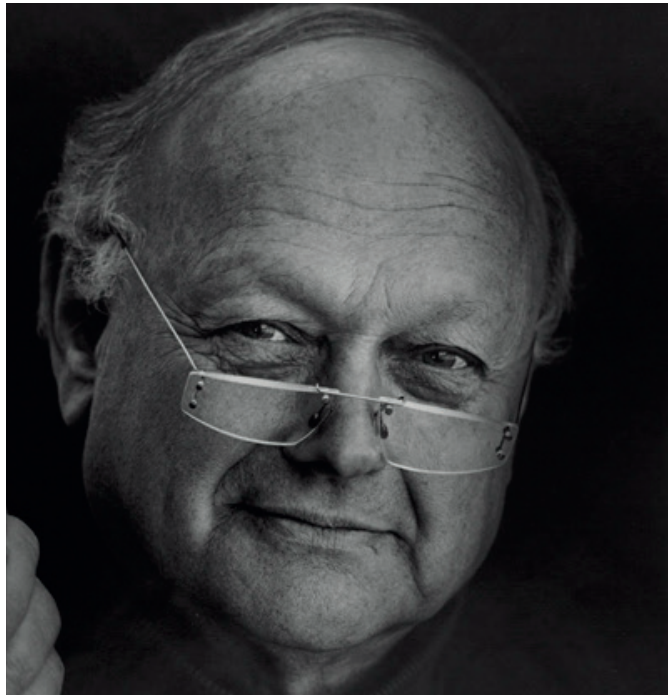


Resedential building for one family

Analysis – Alderton house

16

1. Glenn Murcutt



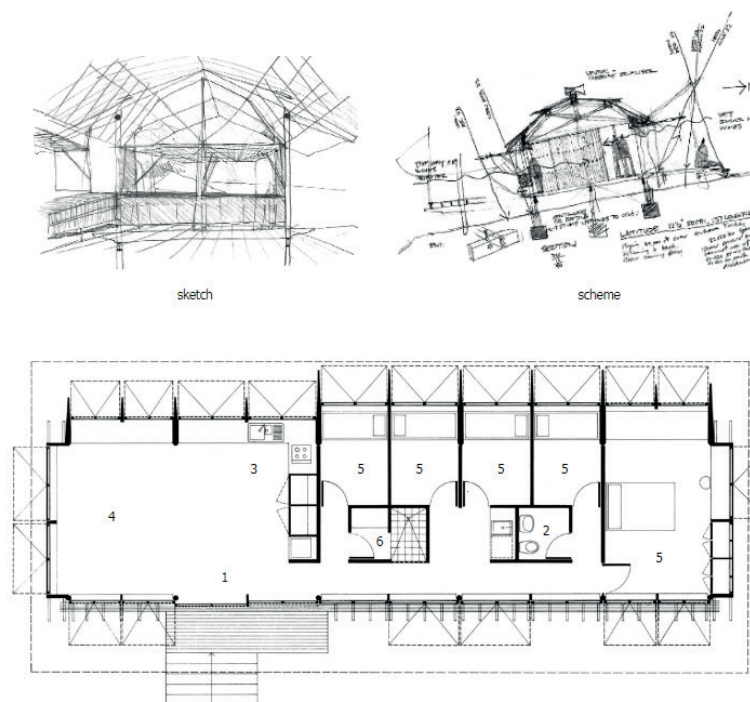
2. Outside view



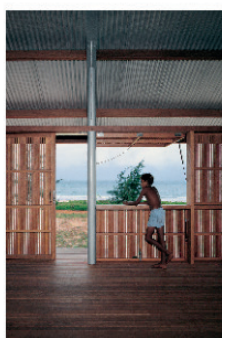
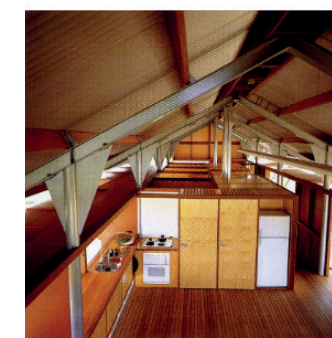
Introduction

The Alderton house was commissioned by Mark Alderton his partner Banduk Marika, who lives in the far north of Australia and is associated with the Marika clan in the area. The project offered a challenge for the design as climatical, and cultural conditions played a very big role here. The area where the house is vested had a tropical climate and the regular cyclones, hard wind, and heavy rainfall with regular flooding's. That's why the building has the wide roof stretching beyond the walls at an angle to keep the hot sun out all year round. The walls themselves are all adaptable, they can open and close letting the air in or keeping the heavy winds out. As for the flooding's the building is also slightly raised from the surface.

3. Plans, sections and construction



4. Inside view



1: Architectuul. [z.d.]. Architects | Architectuul. Geraadpleegd op 5 november 2021, van http://architectuul.com/architects/view_image/glenn-murcutt/780

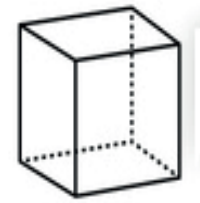
2 3 en 4: Gegeven tekeningen

Tekst: Marika-Alderton House by Glenn Murcutt [630AR] — Atlas of Places. [2019]. Atlas of Places. Geraadpleegd op 5 november 2021, van <https://www.atlasof-places.com/architecture/marika-alderton-house/>

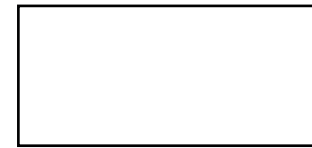
Analysis – Alderton house

17

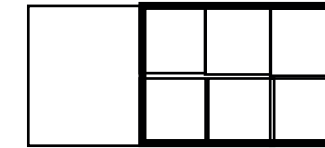
Form & space



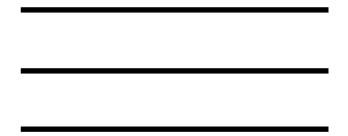
General shapes



Rectangular plan

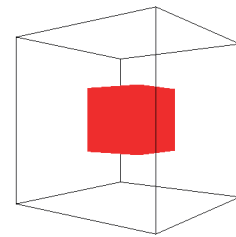


Smaller rooms vs
big open living room

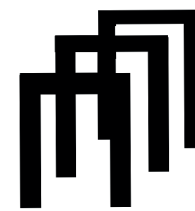


Stretched out
horizontally

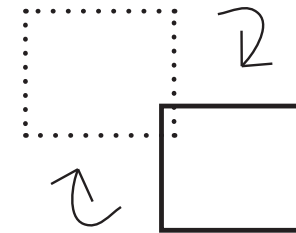
Materialisation & construction



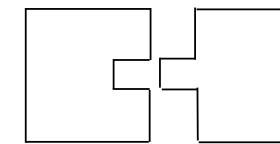
Steel frame work and
local wooden 'skin'



Construction



Adaptable facade



Prefab and bolted to-
gether

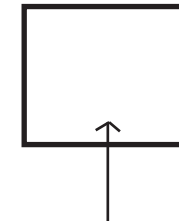


Easily transported

Functionality

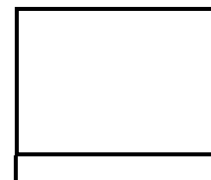


Closed floorplan



One entrance

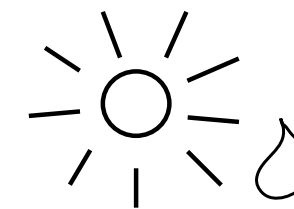
Urban context



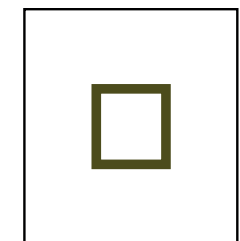
Slightly elevated



Green and open



Harsh weather conditions
Tropical weather



Placed in an open
space with lots of
green

Social context



Community



Residential building for one family
and people from around the area

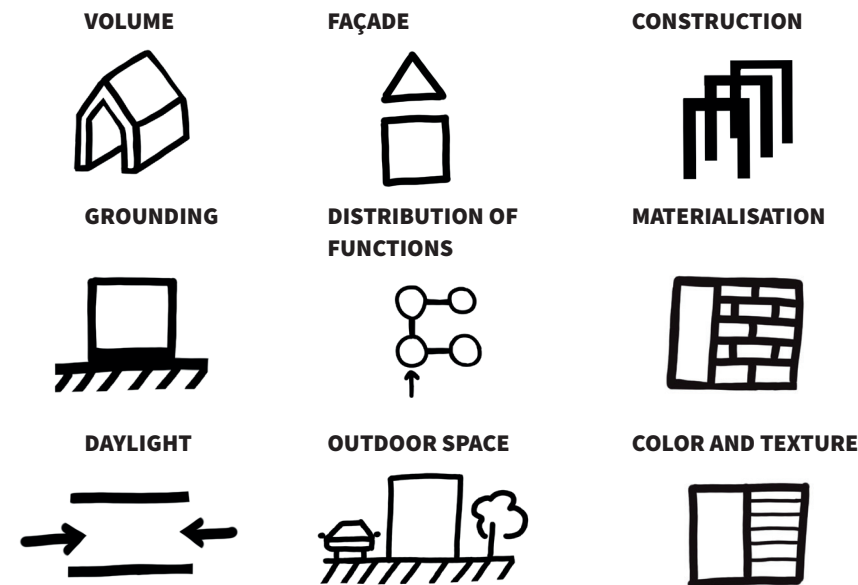
Assignment 1 – Tiny house – ‘Triple roof’

18

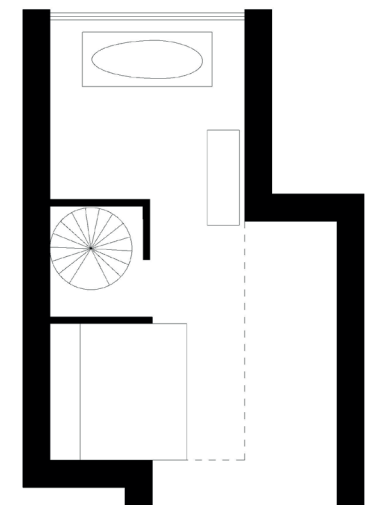
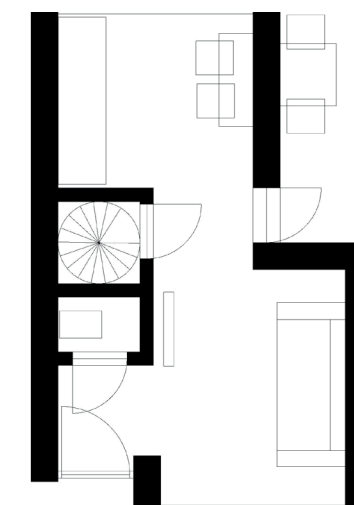
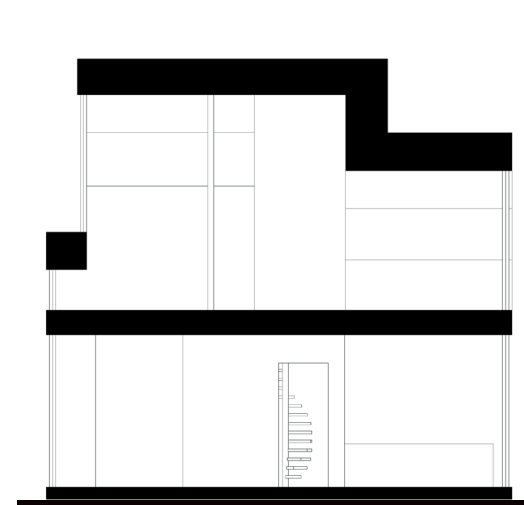
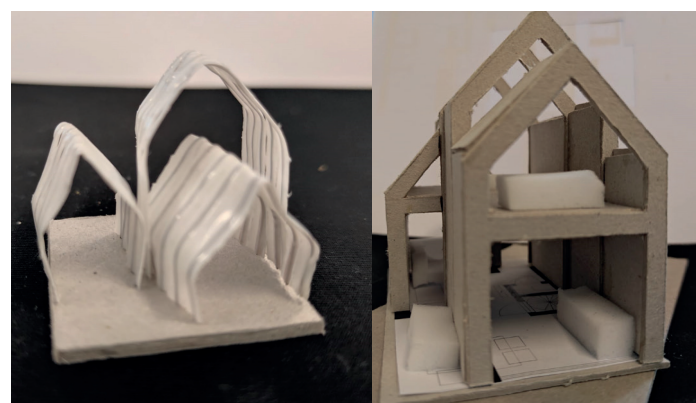
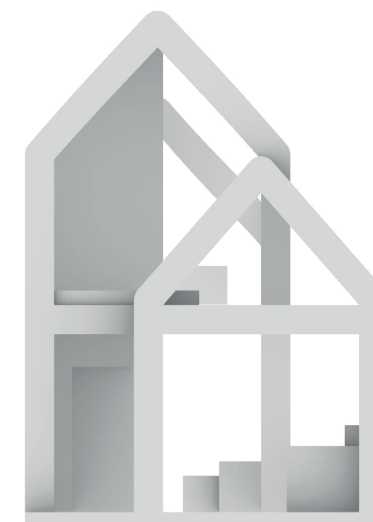
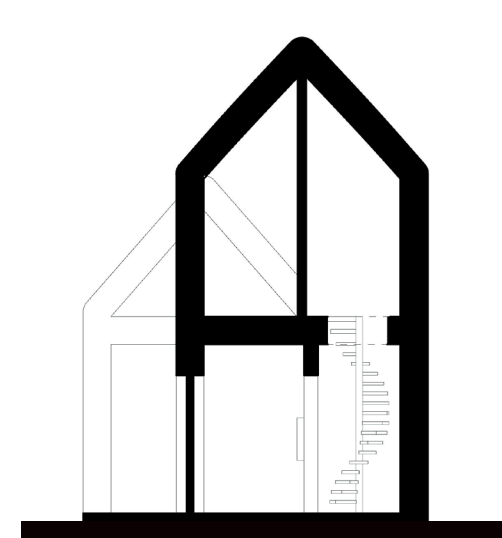
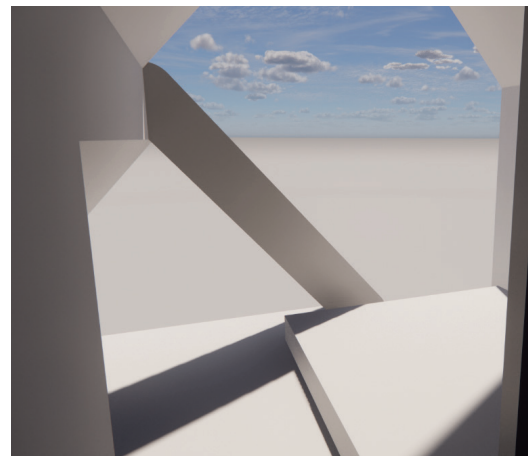
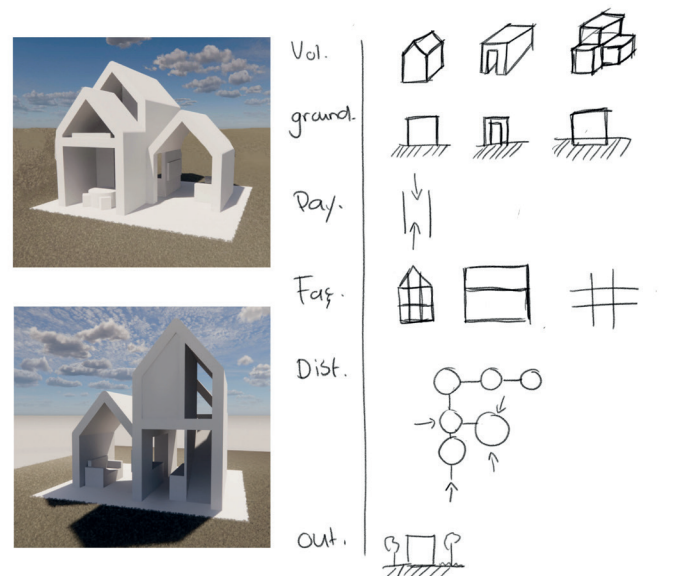
Introduction

The first assignment was to create a tiny house in an area of 7x7x7 meters in any context you wanted it to be. The main concept I started out with was the simple form of a house and started to play around with it until I found a shape I liked and added the program in. Another big feature was the tunnel or binocular like structure to be able to get most of the view outside, which in the instance was situated in a forest area. Later the site changed to a little plot in Voorburg which was less wide and longer, which meant the concept had to adapt a bit to the new situation.

Icons



3D beeld op locatie (Voorburg)



Process

Inside view

Sections

Floorplans and materialisation

Assignment 1 – Tiny house – ‘Triple roof’

19

Domains

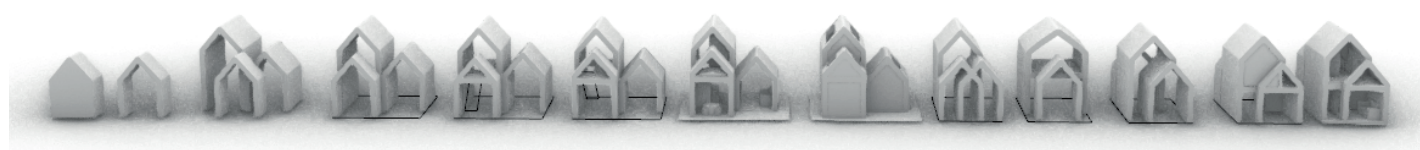
The domains were quite familiar to me however they had never really been specified like they had been in this assignment helped the process of creating a complete design. What was also quite refreshing was the fact we started out more so with the domain of form instead of the urban context or site. You first started out by experimenting with form and function to later adapt the design to the context that was given. One point of improvement for this or next assignments as for the domains is to also focus more on other domains since in this project I focused more on shape and the urban-context and very little on materialization.

Experiment

One of the other aspects that could use some improvement was the experimenting. I noticed that I quite easily went back to old habits and chose one concept and or design and stuck with it as I detailed that one further but did not experiment to much with other concepts or ideas. This could have benefitted the design and made it bolder and extravagant or could have given me other solutions to the same problems who might have worked better for the design.

Laboratory

For the beginning process of finding a nice concept, form, or general idea of a first design I like to use Rhino (a 3D modelling software) to start out making shapes and simple models to get idea's since I think those are quite easily adaptable and you can add changes to your model and keep the old one as well to see if you like the change or not. For this assignment I also made two little physical models. One smaller scaleless one just to get the idea of the shapes and how I was going to arrange them and one 1:50 one to get an idea of the more detailed product. All the other drawings like the floorplan and sections I also drew using Rhino, however the initial idea and arrangement of spaces I drew on paper in a notebook but quickly went on to the digital format. Though like I already said Making littler physical models and or different first sketches for different concepts, shapes etc. would have been nice to have.



Laboratory

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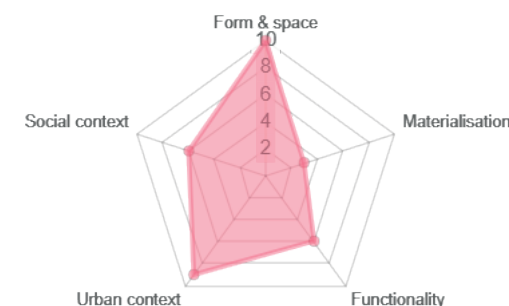
Frame of reference

In the first assignment I didn't really use much reference other than the general shape of houses and to find out certain dimensions of the spaces inside the house. So, for future projects it might also be nice to get a bigger or more elaborate frame of reference also to get more ideas and different ideas for a design.

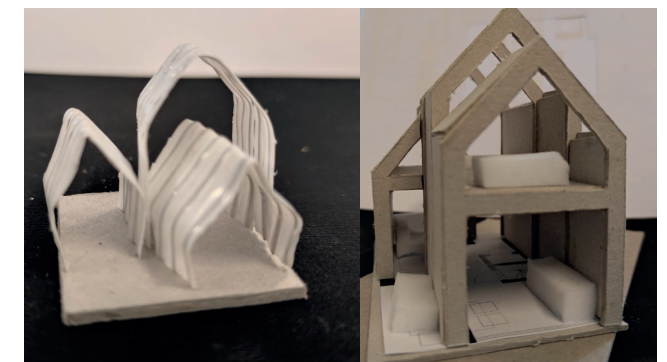
Guiding theme

From the very start the guiding theme was the 'house like shape' like everyone would draw a house if asked. The sloped roof and a box underneath and started to play with that concept. One other guiding theme was the outside view and to connect the outside with the inside. That last one later changed some bit at the site changed as well and I created a more closed side towards the other buildings and an open side to the view towards the church.

Focus of the domains:



Fysical models:



Assignment 2 – CNC Chair – ‘Simply-a-chair’

20

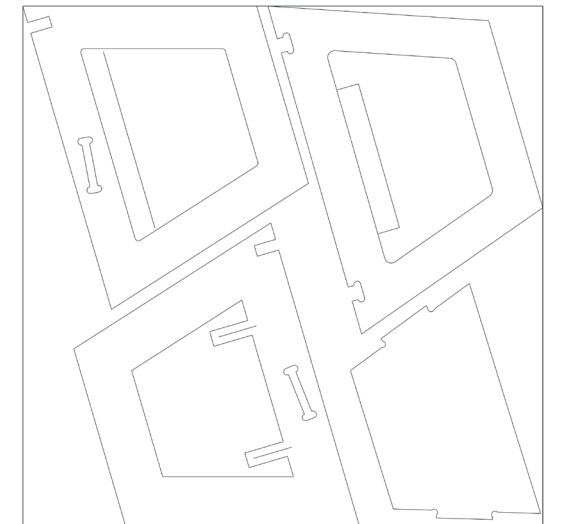
Introduction

The next assignment was the CNC chair, in which we collaborated with TNM (The New Makers) to create a 1:1 prototype of your own chair design. The material was chosen for us, we would get an 18 x1220 x 1220-millimeter slab of wood and from that we had to CNC cut our design to create ‘something to sit on’ without the use of screws, glue, or other additives to connect the pieces (apart from the six cabinets). In other words, everything had to be either ‘friction fit’ or fit so that the pieces together would form a strong enough connection to be able to sit on it. The final main concepts of the chair were the smart use of the limited material (form and contra form), the use of only one shape and it being a chair to quietly read, sit back and relax in.

Process:



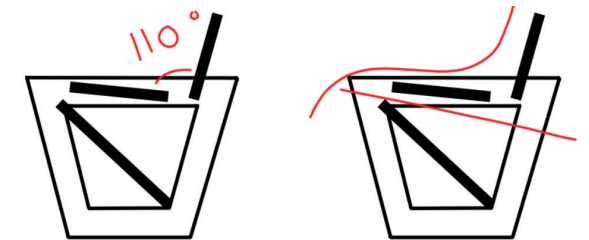
CNC Format/ nesting



Form + contra form



Ergonomics



Ideal backseat angle 100-110 degrees
& Slanted seat for 'lounge' effect

Connections



Assignment 1 – CNC chair – ‘Simply-a-chair’

21

Domains

The domains that I wanted to explore more from the previous assignment was the materialization. Even though the material was already chosen for us as such we didn't really have to explore different materials but more the different characteristics of one material in this case wood. What also jumped out was that again the form and space were quite major aspects in this assignment or at least I focused on that domain the most again, however I explored the other domains more as well to get a more leveled picture.

Experiment

In the previous assignment I hadn't explored to many different concepts so this time I tried to experiment with a lot more diverse options and connections to find a design I would like. I began by drawing different connections that did not need any screws or anything to hold it together but for the material itself and then hopped over to Rhino to create different chairs with those connections I found and started experimenting on the form. Eventually I got a chair I liked and adapted it to fit within the square slab of wood using the form and contra form concept and played around with the placement and sizes of the shapes to land on the prototype you saw on the previous page.

Laboratory

Again, I used rhino to explore the different possibilities to get to a design I like but this time explored more options with the physical models as well as hand drawn sketches. Though Rhino was still the most used medium to get to the first prototype making dozens of variations of a certain concept and trying out what worked for me and what didn't. What was new to me was the use of the CNC cutter to create a 1:1 scale model. I was already familiar with rhino so creating the file for the machine wasn't too much trouble, but I did notice you had to be very precise in the measurements and line types for the different drill functions (outer, inner, drill, etc.).

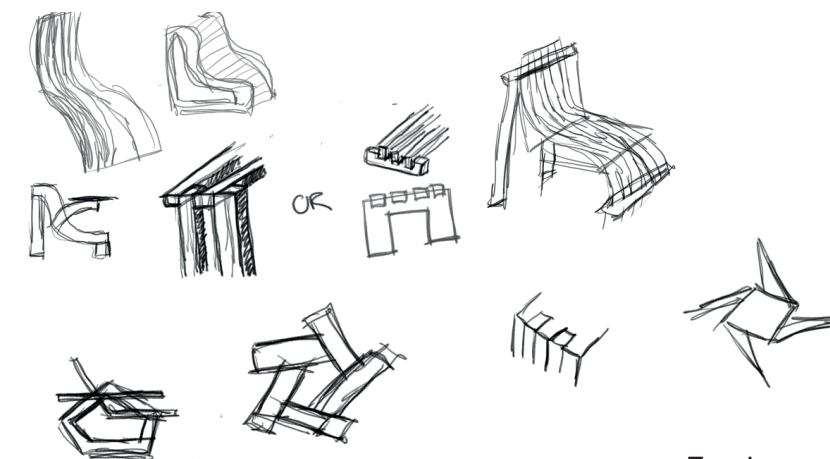
Frame of reference

For this assignment I mostly searched for references for connections that were friction fit and to find the optimal dimensions and angles to create a comfortable chair. What was interesting to see is that while searching you stumbled upon a only few different types of connections that didn't need anything and yet many different designs using the same concepts and yet looking totally different.

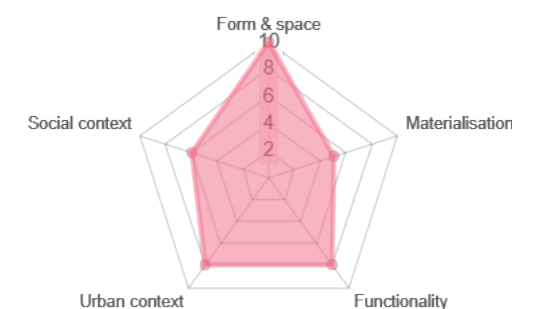
Guiding theme

After having searched numerous different connections and tried out different designs I landed on one concept I favored from the others and that was the use of one shape and repeating that over and over to create the chair. Furthermore, since we only had a limited amount of material you needed to be very careful with where you place, what kind and what size shapes you wanted which led me to the concept of form and contra form. Only using trapeziums and using the outer band for the legs of the chair and the middle to create the seat and back rest. I liked that I first experimented before landing on a solid theme to be able to get the design I wanted.

Sketches of connections and or chairs



Explored domains



Form and contra form



References of connections and different chairs



Assignment 3 – Adaptive house – The windchime

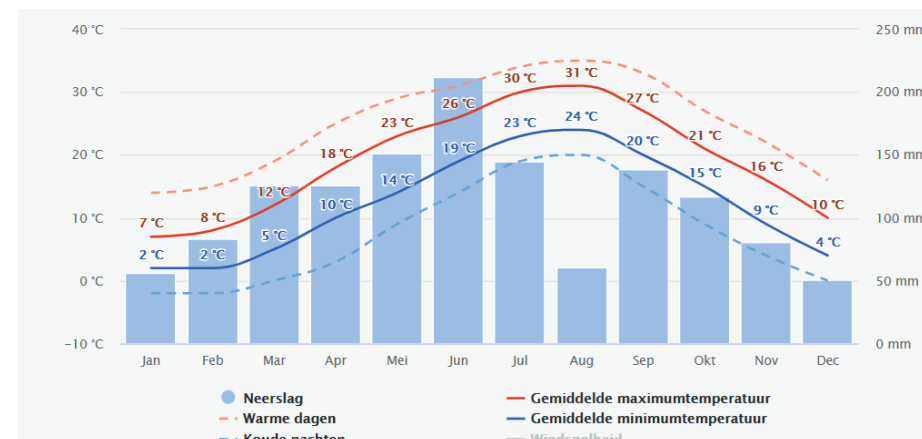
22

Introduction

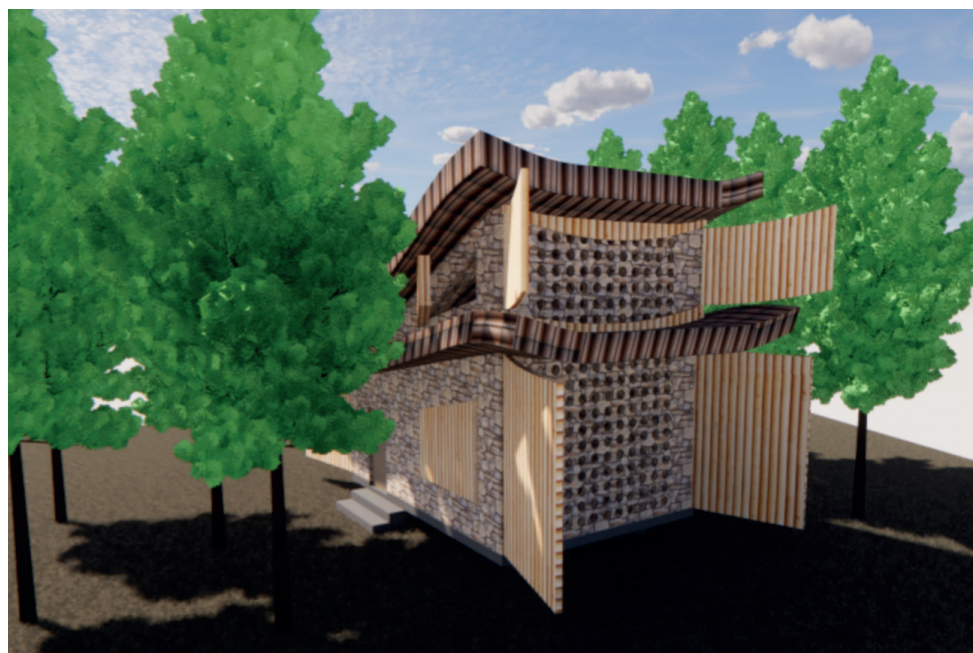
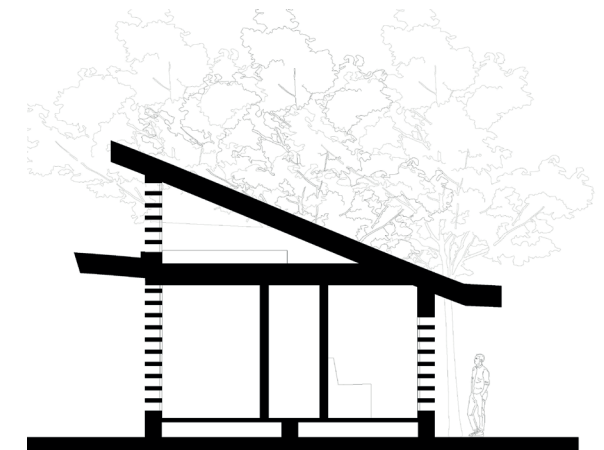
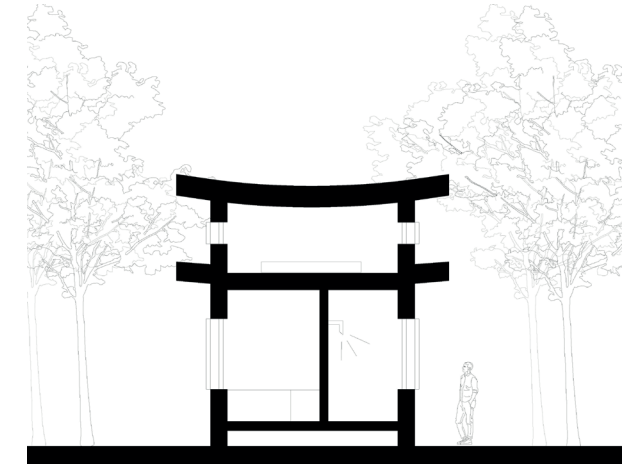
The last assignment was the climate adaptive house. The house had to be around 30 square meters and be able to adapt itself to the changing climate in your chosen area and test the adaptable part of the house yourself to see how much effect it would have. The area I landed in was Osaka Japan. Their climate is very warm and humid in the summer and quite cold and dry in the winter without much difference of night and day temperature. The Windchime, gaining its name by its main function: ventilation, has two porous walls through which air can flow through in the hot summers to cool the rooms down and with the 'cool' bricks that hold on to water they cool the house down even more. In the winter you can close the isolated panels in front of these walls and you'll be able to sit nice and cozy inside without freezing. The shape of the building will also help the air flow through creating the unique shape. Furthermore, the double extended roof creates shadows for in the summer to keep the hot sun out and the tiles are glazed to keep the rays of sun out and the summer green trees provide shade in the summer as well.

Cool bricks

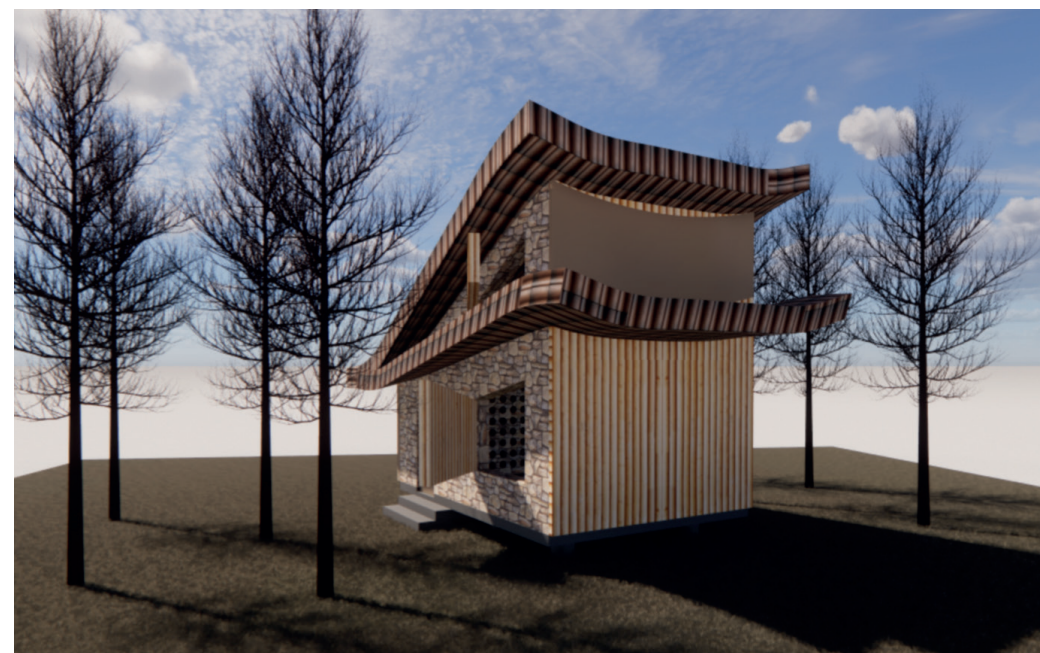
DiStasio, C. (2015b, maart 9). 3D-printed "Cool Brick" cools a room using only water. Inhabitat - Green Design, Innovation, Architecture, Green Building. Geraadpleegd op 7 november 2021, van <https://inhabitat.com/3d-printed-cool-brick-cools-a-room-using-only-water/>



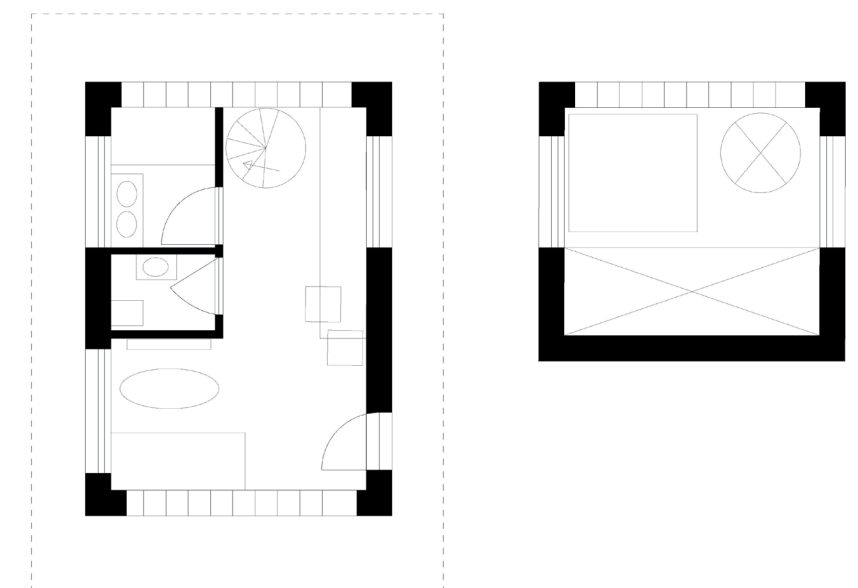
Rainfall and temperature Osaka



Summer situation



Winter situation



Plans and sections

Assignment 3 – Adaptive house – The windchime

23

Domains

In the third assignment I wanted to focus more on the dimensions I hadn't really explored and decided to mostly look at materials to find something that could make the inside environment of the tiny house a lot more comfortable in extreme environments. And since it depended on climate of the area you chose the urban context was also one of the more important dimensions and focused on that as well and less on form and space.

Experiment

The experiments were mostly focused on one aspect of the house to see how your concept would work practically. I found some inspiration in the cool bricks, by Emerging Objects who were able to hold on to water and with their porous structure let the air through to be able to cool the area down through evaporation. I tried to recreate that same concept with 'Oase' (green blocks that are able to take in a lot of water) and poked holes through them to see how much of a difference the wet compared to the dry blocks would work. The wet ones had more effect and cooled the area down by 4 degrees in ten minutes instead of just 2 degrees in 10 minutes. I also tried out different sized holes to see if that would make a difference, but I couldn't see much of an effect there. You could also notice the temperature would faster duck under the outside temperature than the dry ones. The only problem was that the wet ones would raise the humidity and in Osaka, you would want to refrain from getting the area more humid or even get the humidity down. So I tried to find a material that would let air through and water out and found one but didn't get to experiment with it yet to see if the system of the cool bricks would still be effective.

Laboratory

In this case I also used rhino for the general shape of the building and to make renders but what was the most interesting was the fact we were able to use things like a thermal heat camera and the hobo (device that can read the temperature out and inside and measure the humidity) to be able to experiment with the climate adaptive features of our houses. I also tried out a new material to make my prototype model in 1:50 scale since it was going to be quite hard to get the fluid shape of the building with just cardboard and decided to use clay since that was much easier to shape but ended up being quite time consuming. Another source of information was climate consultant which provided detailed information about the climate of an area of your choice, and since I had already worked with the program it was quite an easy way to gain information on that aspect. However the website Meteoblue was much clearer and easier to understand than the consultant.

Frame of reference

One of the biggest references was the cool bricks by Emerging Object. That reference created the concept of my building and from that I began to adapt it to something that would fit in the climate I had chose my house to be. Additionally, I tried to find as many techniques as possible that were used in various other architectural projects to find the ones that would be fitting and was amazed by the simplicity and yet effectiveness of a lot of techniques since I was used to use or add quite complicated systems or techniques to keep a space comfortable even though sometimes the solution can be much simpler.

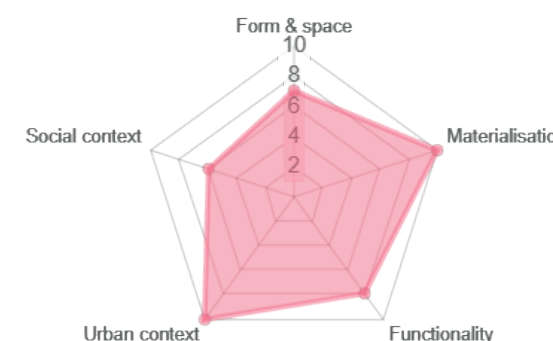
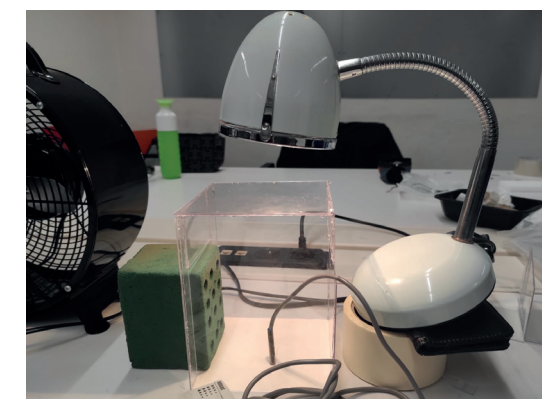
Guiding theme

As said before the main aspect and guiding theme was for me the bricks that would cool down a room by combining ventilation an evaporation. The biggest guiding theme in the project though was the adaptability or the solution to the humid and hot summer and cold and dry winter in Osaka to create a comfortable inside climate. And by focusing on the bricks, I had a hard time focusing on the humid environment of the area and tried to find a solution quite late so I wasn't able to fully explore that aspect to see if it would be as effective as believed to be.

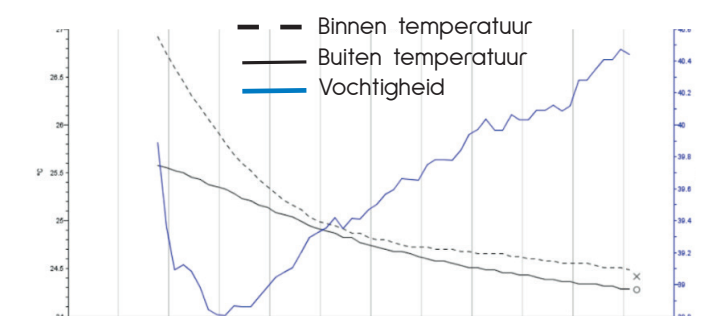
Air filter: TEMISH



Setup experiment



Explored domains



Results of HOBO:

Reflection Archineering 1

At the start of the minor, I was surprised as to how much different the view was on the different exercises and assignments on the buildings as to how I have been taught in the years before of my study. I study architecture but there you mostly start with a program a site and different rules or restrictions. Here we started with a project that had a free or choice site a very simple program and you were guided to explore and see for yourself where the boundaries are at and focus more on the designing instead of just taking one idea and work that out in detail.

Another skill gained from the minor is the ability to form a nice looking and complete concept for a building or other projects within a small amount of time. In two weeks, we were able to design a chair and make a functioning 1:1 prototype. And in just under three weeks being able to create a plan and sections and concept for a climate adaptive house. It might not be perfect or detailed yet, but it does help get the message over to the people.

Furthermore, what I also liked is the fact that for example on the first assignment we were freely designing something and only later added the restrictions so you could design a more extravagant and bolder house and see how you could adapt it to a different site and what the consequences of that would be.

I gained many new tools to use for my future projects and hope to keep the experimenting and fast way of setting up a concept through those experiments with me to the next project.

